UNIVERSITY OF SRI JAYEWARDENEPURA - FACULTY OF APPLIED SCIENCES

BSc General Degree Second Year First Semester Course Unit Examination – September 2023

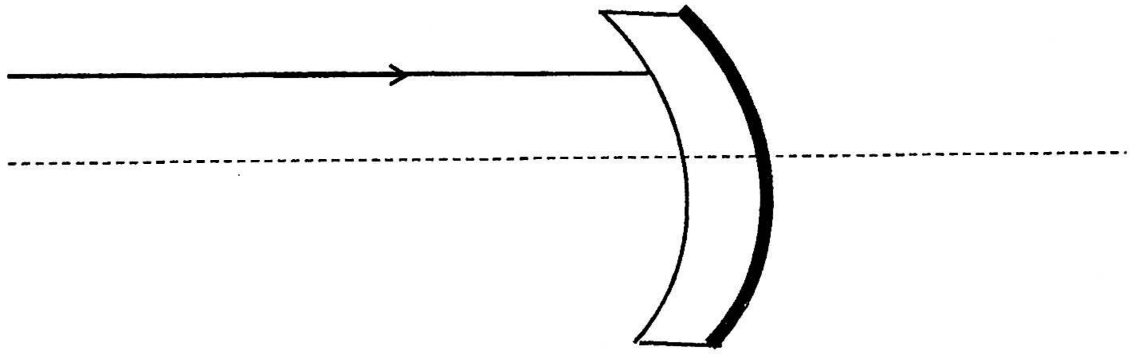
DEPARTMENT OF PHYSICS

PHY201 2.0 Optics 1

Time: Two (02) hours **No. of questions:** 04 **No. of pages:** 04 **Total marks: 100**

Answer ALL questions

1) The optical accessories known as the thick mirrors are made by applying a reflective coating on the back surface of a thick transparent material such as glass. Shown in the figure below is a sketch of a thick mirror, made out of glass (refractive index $n = 1.5$) and the back surface is fully silvered. This thick mirror's radius of curvature is 10.0 cm and the thickness is 3.0 cm. A light ray is incident on the front plane, parallel and 5.0 cm above the main axis.



i. Using proper diagrams, **briefly define** the primary focal point for a concave spherical refracting surface and for a concave reflecting surface. (04 marks)

ii. If the above is considered as a thin mirror (silver coating on the front plane) **find** its theoretical value for focal length. (01 marks)

iii. If you consider only the thick curved glass (disregarding the reflecting coating(s))
   (a) **Find** the theoretical value for its secondary focal length? (07 marks)
   (b) **Is this** a diverging, converging or neutral system? **Explain.** (03 marks)

iv. **Draw** the following on separate diagrams using a suitable scale. *Show all your construction lines for full marks.* [Recommend 1:1 scale for the ray diagram & 1 = 4 cm (or larger) scale for refractive index in the constructions. State the scale on the diagrams, if using a one different to above]. *Note: providing the information within 15-20 cm span around each optical accessory is sufficient.*
   (a) **The path of the ray** for the thin mirror (*accessory* in ***part ii***). (05 marks)
   (b) **The path of the ray** for the curved glass (*accessory* in ***part iii***). (05 marks)
   (c) **The path of the ray** for the thick mirror (back silvered curved glass). (05 marks)

v. Can the focal length found in **part *ii or iii*** be different from the value found using the ray diagram in ***part iv (a)*** or ***part iv (b)***. **Explain** your answer. (*Assume the diagram have no construction errors*). (05 marks)

[35 marks]

2) When two sets of waves are made to cross each other, at the region of crossing there are places where the disturbance is practically zero and others where it is greater than that given by either wave alone. This is known as the principle of superposition.

i. **Write** the general formula for a sinusoidal wave with angular frequency ω and wave vector $\boldsymbol{k}$ in to the direction of propagation (+x). (02 marks)

ii. **Derive** the formula for a wave produced by superposition of two waves of equal frequency (ω) and amplitude ($\boldsymbol{a}$) traveling in the same direction (+x), but one with a distance Δ ahead of the other. (06 marks)

iii. **Express** your analysis regarding the amplitude of the new wave. (03 marks)

iv. Two waves traveling together along the same line are represented by

$$y_1 = 25 \sin(\omega t - \pi/4) \quad \text{and} \quad y_2 = 15 \sin(\omega t - \pi/6)$$

(a) **Find** the resultant amplitude (02 marks)

(b) **Find** the initial phase angle of the resultant (02 marks)

[15 marks]

3) Coherent light sources or diffraction of light passing through slit(s) generate light patterns on a screen due to interference. The regular used slits have a very large length compared to its width.

i. **Briefly explain** "coherent light sources". (02 marks)

ii. **Briefly define** the conditions defining Fraunhofer and Fresnel diffraction. (04 marks)

iii. Using a proper diagram, **briefly explain** how a single slit generate the diffraction pattern (05 marks)

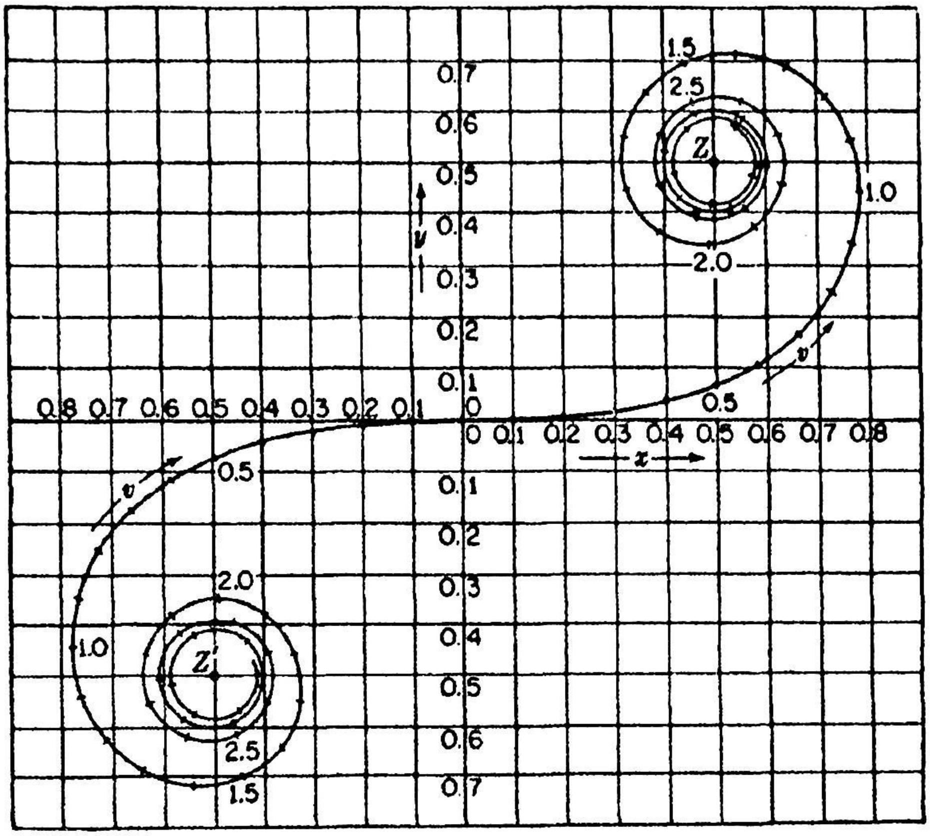
iv. Assuming the slit width is $\boldsymbol{b}$, the distance from the center (O) of the slit to any arbitrary point (P) on the screen is $\boldsymbol{x}$, and Θ as the angle between the center axis and OP. **Write** the <u>complete equation</u> for the interference due to <u>two partial sources</u>. The source parameters are frequency ω and amplitude $\boldsymbol{a}$. (05 marks)

v. **Write** the formula for <u>the intensity</u> of Fraunhofer Diffraction pattern for a <u>regular slit</u> (used in <u>*part iv)*</u> **and** for <u>a rectangular slit</u>. **Identify** all the parameters. (03 marks)

vi. **Express** your analysis regarding the diffraction patterns for a regular slit compared to a regular slit. (04 marks)

vii. **Define** Half-Period Strip (02 marks)

viii. Use the Cornu's spiral to **evaluate** the following. (***Note:*** *reading the data to <u>first decimal place</u> is sufficient*)



(a) Intensity due to first half period stripe in upper region (02 marks)

(b) Intensity due to first half period stripe. (01 marks)

(c) Intensity due to full wave front. (02 marks)

[30 marks]

4) Light can be polarized using both reflection and refraction.

i. Malus and Brewster's did some studies regarding the polarization by reflection.

(a) **What** is/are the discovery(s) lead to Brewster's law? (02 marks)

(b) Using a proper diagram illustrating the conditions, **derive** the Brewster's Law. (04 marks)

ii. Nicol Prism uses double refraction to polarize light,

(a) **Briefly explain** what "double refraction" is. (02 marks)

(b) **Name** the ray types generated in double refraction and define their differences. (04 marks)

(c) **Draw and briefly explain** the Nicol Prism. (04 marks)

iii. **Explain** two other methods used or can be used to polarize light. (04 marks)

[20 marks]

**************************** ***End of the Question Paper*** *********************

Few equations/formulas that may be useful

$$y = 2a\cos\frac{k\Delta}{2}\sin\left[\omega t - k\left(x + \frac{\Delta}{2}\right)\right]$$

$$A = atrt'\frac{1}{1 - r^2}$$

$$I \approx b^2 l^2 \frac{\sin^2\beta}{\beta^2}\frac{\sin^2\gamma}{\gamma^2}$$

$$\frac{n}{s} + \frac{n'}{s'} = \frac{n' - n}{r}$$

$$\beta = \frac{\pi b(\sin i + \sin\theta)}{\lambda}$$

$$\frac{n}{s} + \frac{n'}{s'_h} = \frac{n' - n}{r} + \left[\frac{h^2 n^2 r}{2f'n'}\left(\frac{1}{s} + \frac{1}{r}\right)^2\left(\frac{1}{r} + \frac{n' - n}{ns}\right)\right]$$

$$I \approx A^2 = 2a^2(1 + \cos\delta) = 4a^2\cos^2\frac{\delta}{2}$$

$$\sin A + \sin B = 2\sin\tfrac{1}{2}(A + B)\cos\tfrac{1}{2}(A - B)$$

$$\tan\theta = \frac{a_1\sin\alpha_1 + a_2\sin\alpha_2}{a_1\cos\alpha_1 + a_2\cos\alpha_2}$$

Gaussian formulas

$$\frac{n}{f} = \frac{n'}{f'_1} + \frac{n''}{f''_2} - \frac{dn''}{f'_1 f''_2} = \frac{n''}{f''}$$

$$A_1F = -f\left(1 - \frac{d}{f'_2}\right)$$

$$A_1H = +f\frac{d}{f'_2}$$

$$A_2F'' = +f''\left(1 - \frac{d}{f'_1}\right)$$

$$A_2H'' = -f''\frac{d}{f'_1}$$

Power formulas

$$P = P_1 + P_2 - \frac{d}{n'}P_1P_2$$

$$A_1F = -\frac{n}{P}\left(1 - \frac{d}{n'}P_2\right)$$

$$A_1H = +\frac{n}{P}\frac{d}{n'}P_2$$

$$A_2F'' = +\frac{n''}{P}\left(1 - \frac{d}{n'}P_1\right)$$

$$A_2H'' = -\frac{n''}{P}\frac{d}{n'}P_1$$